

GRID INFRASTRUCTURE AS A BINDING CONSTRAINT ON RENEWABLE ENERGY DEPLOYMENT

A Comparative Institutional Analysis of Germany and Great Britain

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Preface

This dissertation began from a simple question that became more complex the more I studied it: if renewable electricity is now increasingly cheap, politically supported and technologically mature, why does deployment still run into such persistent limits?

My academic background is mainly in economics, political science and data science,, with a particular interest in how markets, regulation and public policy shape real-world outcomes. Through microeconomics and industrial organisation, I became interested in how incentives determine where firms invest, how infrastructure is priced, and how market design can create unintended distortions. Through macroeconomics and energy-related modules, I became more aware of the wider importance of the energy transition for growth, inflation, industrial competitiveness and energy security. My interests in sustainability and clean energy development also exposed me to the finance of renewable generation, project economics and the investment case for clean infrastructure.

However, much of my earlier understanding of the energy transition was generation-focused. I had looked more closely at renewable project finance, investment incentives and decarbonisation targets than at the less visible systems that make renewable deployment possible: transmission grids, electricity-market design, balancing mechanisms, planning consent and the social acceptance of infrastructure. Over time, it became clear to me that the main question was not simply whether countries could build more wind and solar, but whether their institutions and networks could integrate that electricity efficiently.

This is why I came to focus on transmission grid constraints. With the guidance of my supervisor, I narrowed the topic to two countries, Great Britain and Germany because they provide a useful comparison. Both are advanced European electricity systems with ambitious decarbonisation goals and large renewable pipelines. Both face a similar physical problem: renewable resources are increasingly concentrated in peripheral regions, while demand remains concentrated elsewhere. Yet they organise their electricity systems differently, with different grid-governance structures, redispatch mechanisms, pricing debates and approaches to socialising costs. This made the dissertation topic interesting.

Abstract

Germany and Great Britain are among Europe's most ambitious renewable-electricity deployers, yet both face a similar integration problem: renewable generation is increasingly located in peripheral regions while demand remains concentrated elsewhere. This dissertation argues that transmission grid infrastructure has become one of the binding physical constraints on renewable electricity deployment in both countries. It compares how two different institutional systems, Great Britain's unified system operator and Germany's four regionally coordinated TSOs, manage the same underlying problem.

Drawing on data and recent market-design decisions including Great Britain's 2025 rejection of zonal pricing, the dissertation applies three analytical lenses: mechanism, distributional consequence and comparative interpretation. It finds that Great Britain and Germany differ in grid organisation, redispatch design and cost recovery, but converge on persistent bottlenecks, rising congestion-management costs and similar delivery timelines for major reinforcements.

The central finding is that institutional architecture shapes the distribution, visibility and attribution of renewable-integration costs more than the speed of physical delivery. British arrangements make balancing and constraint costs more explicit and politically attributable, while German redispatch, network-charge and technical-standard mechanisms diffuse costs through regulated tariffs and project requirements. However, both systems remain constrained by planning consent, local opposition, HVDC supply chains and skilled-labour limits. The dissertation concludes that transmission expansion remains essential, but cannot be the only policy response. Demand-side flexibility, storage, locational signals and grid-code requirements are complements to grid build, and the frontier policy question is how to combine them while keeping costs politically durable

Part I — Framing the problem

1. Introduction

By the time connections reform was implemented in 2025, the grid had become a binding bottleneck on Britain's renewable build-out: the old connections queue had grown to over 700 GW of generation, storage and demand projects, more than three times the 200–225 GW of clean generation capacity identified as necessary by 2030, with some shovel-ready projects facing connection dates up to a decade away (Ofgem, 2024a; NESO, 2024d; NESO, 2025a; DESNZ, 2024). Pressure is now compounding from the demand side: the GB transmission queue for new demand connections rose from roughly 17 GW in November 2024 to 97 GW by June 2025, driven notably by data-centre applications and broader electrification (Ofgem, 2025a). Germany confronts the mirror problem. Of the roughly 16,800 km of transmission lines legally identified as necessary for grid expansion, only around 4,700 km had completed planning approval or no longer required it by the end of 2025, even after a record 2,000 km of power-line approvals in 2025 alone (Bundesnetzagentur, 2026a). In both systems, generation and electrification are outpacing the wires needed to connect them.

The integration of renewable electricity sits at the intersection of three policy imperatives that have hardened over the past decade. First, net-zero targets, which are represented by UK's Climate Change Act 2008 commitment to 2050 and Germany's Klimaschutzgesetz obligation to 2045, are legally binding and presuppose near-complete electricity decarbonisation (HM Government, 2019; Bundesregierung, 2021; DESNZ, 2024; BMWK, n.d.). Second, energy security concerns, which have sharpened since post-2022 supply disruptions and have elevated domestic renewable generation from a climate policy to a strategic priority for energy independence (IEA (2022a); IEA (2022b)). Third, industrial competitiveness, which increasingly depends on access to affordable, reliable and low-carbon power, as energy-intensive industries face high electricity costs and decarbonisation pressures (Draghi, 2024; European Commission, 2025). Each agenda depends on the same physical and institutional constraints: generation and storage capacity, the transmission and distribution networks that connect them to demand, the planning and consent regimes that authorise build-out, the supply chains and skilled labour that deliver it, and the market and regulatory design that allocates costs and risks. Each of these layers can independently bind.

This dissertation focuses on the transmission grid infrastructure as a binding physical constraint on renewable electricity integration in Great Britain and Germany. Three claims

follow. First, the constraint is spatially and economically concentrated, especially for wind, which makes the "build more wires" debate a question about specific corridors. Second, the comparison between GB's centralised single-TSO system and Germany's four federally-coordinated TSOs reveals symmetrical delivery timelines produced by different institutional pathologies, suggesting that institutional design shapes the character of the delay more than its scale. Third, because supply-side acceleration appears to have hit consent and manufacturing limits, demand-side instruments such as flexible load and storage become particularly central among the complements to grid expansion.

The research question is therefore both empirical and comparative: why do Great Britain and Germany, despite legally binding decarbonisation targets and historically large transmission investment programmes, continue to curtail wind at scale, incur multi-billion annual constraint costs, and project these bottlenecks to persist through the late 2020s? And why do two systems with markedly different institutional architectures converge on similar commissioning timelines and a similar trajectory on costs?

Those architectures differ on three dimensions relevant to grid delivery. Great Britain operates with a single, recently-independent system operator separated from transmission asset ownership, whereas Germany operates with four regionally-coordinated TSOs that own the assets they plan. Consent in Great Britain follows a centralised single-stage Development Consent Order regime under the Planning Act 2008, while Germany combines federal *Bundesfachplanung* with Land-level *Planfeststellung*, with parallel approvals for offshore connections (BNetzA, 2025; HM Government (2008b); MHCLG/DLUHC (2024)). And cost recovery is partially socialised through visible BSUoS charges in Great Britain, but largely embedded in regulated network tariffs (Netzentgelte) in Germany.

This dissertation argues, on the basis of the boundary-stress evidence in Part I and the comparative institutional analysis developed in Parts II–III, that institutional architecture in Great Britain and Germany shapes how renewable integration costs are generated, distributed and contested: who pays, who decides, and who is exposed to backlash. It does not, however, appear to fully determine the pace of physical delivery. Delivery timelines are also constrained at the operational margin by HVDC converter-station supply chains, multi-stage planning consent, skilled construction labour and local public acceptance, all of which lie partly outside what national institutional reform can quickly change (IEA, 2025; Bundesnetzagentur, 2023; BMW, n.d.). The policy implication matters because supply-side institutional reform, restructuring system-operation roles, accelerating grid connections and

consent regimes, and tightening regulatory delivery incentives, has been positioned as a principal lever for accelerating renewable build-out in both jurisdictions (DESNZ, 2024; Ofgem, 2023; Bundesnetzagentur, 2023; BMW, n.d.). If the marginal returns of these reforms on commissioning timelines are limited, then the question becomes how each institutional architecture can adapt around the constraints it cannot quickly remove.

2. Literature, framework and approach

The dissertation draws on four bodies of literature. First, the economics of transmission investment, where Joskow and Tirole (2005) and Joskow (2007) establish that transmission grids are classic natural monopolies in which merchant-investment incentives are insufficient and some form of regulated or coordinated investment is required. Second, the literature on locational pricing, where Hogan (1992), Pollitt (2023), and Eicke and Schittekatte (2022) argue that nodal or zonal markets can deliver efficiency gains relative to uniform-price zones in systems with significant structural congestion. Third, the emerging grid-focused policy literature typified by the IEA (2023) stocktake and ACER reporting on European electricity security and system integration (ACER, 2024). Fourth, the literature on the political economy of energy transitions, which draws attention to the distributional consequences of infrastructure decisions and to how security-of-supply or system-stability arguments can become entangled with path-dependent institutional interests (Monopolkommission, 2023; Meadowcroft, 2009).

The analytical framework rests on three lenses applied to each substantive chapter in Part III. The technical lens asks how the grid physically works and where its limits lie. The economic lens asks how costs are generated and who bears them under current price and tariff design. The political economy lens asks how authority over decisions, the visibility of their costs, and the organisation of affected interests determine which reforms are politically feasible. The approach is interdisciplinary because each lens taken alone produces an incomplete account: for example, a purely technical treatment cannot explain why projects identified as necessary in 2010 commission in 2028 rather than 2018; a purely economic treatment cannot explain why Germany resisted splitting its single bidding zone despite the 2025 ENTSO-E Bidding Zone Review finding that a Germany–Luxembourg split into five bidding zones produced the highest positive efficiency result among the analysed German configurations, or why Great Britain rejected zonal pricing in 2025 despite a theoretically compelling case (ENTSO-E, 2025; Clean Energy Wire, 2025a; DESNZ, 2025). Within each chapter, the three lenses are applied through three analytical moves: a mechanism move that describes how an

institutional or technical feature physically produces an outcome, a distributional move that identifies who wins, who loses, who pays whom and who decides, and a comparative move that tests whether Great Britain and Germany converge or diverge on that outcome and asks what the comparison implies about the explanatory weight of institutional design itself.

Methodologically, the dissertation is a comparative institutional analysis. The core method is qualitative and conceptual, examining supply-side reform, market design, ownership and governance across the two jurisdictions, alongside factors exogenous to how institutions regulate Transmission Systems but which impact them: supply-chain disruptions, social and political constraints, and demand-side responses. This is supported by a quantitative empirical component for Great Britain that provides geographical evidence of where the constraint physically binds, and by structured indicator comparison for Germany. The primary data sources are NESO balancing-cost publications and monthly constraint breakdowns for Great Britain (NESO, 2024a; NESO, 2025e), Bundesnetzagentur monthly redispatch publications and SMARD transparency data for Germany (Bundesnetzagentur, 2024; Bundesnetzagentur, 2025a; SMARD, 2025), and the published investment plans Beyond 2030 (NESO, 2024b) and the Netzentwicklungsplan 2037/2045 (Netzentwicklungsplan, 2025).

The empirical component is an original boundary-stress analysis for Great Britain, combining NESO's ETYS March 2025 GIS shapefile with the Day Ahead Constraint Flows and Limits dataset, which records day-ahead limits and flows at relevant boundaries (NESO, 2025f; NESO, 2025g). The primary indicator is the ratio of observed flow to binding capacity limit, from which I derived mean stress, near-saturation frequency and a spatial mapping onto ETYS boundary polygons. This is supplemented by structured indicator comparison for Germany using redispatch volumes by federal state and negative-price hours indicators. The temporal focus is 2019–2025, a period over which constraint and redispatch volumes have grown sharply in both systems and over which institutional responses have accelerated.

By focusing mainly on transmission grid capacity and its institutional governance, the dissertation leaves several adjacent factors outside its scope, each of which interacts with the constraints analysed here and contributes to renewable deployment delays, sometimes to a comparable degree. Three are worth flagging. First, the analysis focuses on transmission, with distribution treated only where it interacts with transmission (curtailment caused by distribution bottlenecks, prosumer integration). Second, it focuses on electricity, with hydrogen and gas treated only where they interact with electrification and electricity-grid

planning. Third, it does not engage with the broader theoretical argument that uniform pricing structurally distorts locational signals (Hogan, 1992; Pollitt, 2023; Eicke and Schittekatte, 2022), nor does it attempt a formal cost-benefit analysis of zonal versus national pricing for Great Britain; it instead characterises the political-economic arguments surrounding the July 2025 REMA decision (DESNZ, 2025).

Part II — The shared physical problem

4. Spatial mismatch and the physics of transmission

4.1 The geography of the binding constraint

Germany and Great Britain share a defining spatial feature: the best renewable resources are peripheral, while demand is central. In Great Britain, Scotland hosts the country's windiest onshore sites and the bulk of its offshore wind pipeline, with over 30 GW of offshore wind planned in Scottish waters by 2035 against only 6 GW of peak Scottish demand, an indication of Scotland's crucial role in supplying electricity to the rest of Great Britain (NESO, 2024e). In Germany, high wind feed-in in the north and north-east must increasingly be transported to consumption centres in the west and south, including Bavaria and Baden-Württemberg (Netzentwicklungsplan, 2023; Bundesnetzagentur, 2026a). In both countries, the transmission system was historically built for a generation pattern dominated by large central and southern thermal plants, rather than peripheral renewable generation.

This peripheral-to-central flow pattern is the direct consequence of wind and solar replacing dispatchable thermal plants whose location was dictated by proximity to industrial demand, fuel logistics and cooling water. In economic terms, the renewable transition has reshaped the spatial distribution of the electricity production function, while the transmission network, a slow-moving capital stock with multi-decade lead times, has lagged behind. The result is structural congestion at specific boundaries: B6 in Great Britain (see section 5.4 for further detail), and the north–south corridors now being addressed by SuedLink, SuedOstLink, A-Nord and Ultranet in Germany (Regen, 2024; Bundesnetzagentur, 2026a).

4.2 The three binding limits on transmission capacity

Transmission capacity is bounded by three physical limits that operate in parallel, any of which can bind first. The thermal limit is the wire's ability to carry current without

overheating, determined by conductor size and climate (temperature and wind cooling). The voltage stability limit is the receiving end's ability to maintain voltage under high real-power flow, determined by the local availability of reactive power. The angular stability limit is the ability of generators at each end to remain synchronised through faults, determined on long AC lines by the electrical distance between them (Kundur, 1994). N-1 security rules, the requirement that the system survives the loss of any single element, mean that operators preserve contingency margins, so available transfer capacity can sit materially below nominal line capacity. The observed utilisation of B6 at 50–65 per cent of nominal capacity in 2024 (Regen, 2024) reflects the gap between nominal and operationally available capacity, including contingency margins, outage conditions and also active constraint management.

Different binding limits imply different solutions. Thermal-limited boundaries can be relieved by Dynamic Line Rating (DLR) sensors, which let operators use more of the thermal envelope in favourable weather. Germany deploys the equivalent, weather-dependent overhead-line operation, or WAFB, and the Netzentwicklungsplan 2037/2045 illustrates potential utilisation rising from 100% to 130% of rated capacity under favourable weather conditions (Netzentwicklungsplan, 2025). Great Britain is also deploying DLR selectively on existing overhead lines to increase safe capacity and reduce constraints (National Grid, 2025). This technology adds meaningful capacity at the margin but remains insufficient where corridors are structurally constrained, because it optimises existing assets rather than changing the geography of transfer capacity. Voltage-limited boundaries are addressed by synchronous condensers, STATCOMs and other reactive-power compensation equipment. Angular-limited boundaries require faster protection systems, series compensation, grid-forming inverters and, for long-distance bulk transfer, HVDC conversion, which removes angular separation as a binding AC constraint (Kundur, 1994; Kirschen and Strbac, 2018; Netzentwicklungsplan, 2025). These distinctions matter because the quality of grid operation is multi-dimensional in both its causes and its solutions. The shorthand of “building more wires” obscures this: relieving a thermal constraint, a voltage-stability constraint and an angular-stability constraint each calls for a different type of investment, and the binding constraint on any given corridor determines what “more grid” actually means in practice.

4.3 Frequency, inertia and grid services

Separately from transmission capacity, the grid must maintain system-wide frequency close to its 50 Hz target. Frequency reflects the real-time balance between supply and demand: when generation falls below demand, frequency falls; when generation exceeds demand,

frequency rises (NESO, 2024f). Synchronous generators: coal, gas, nuclear and hydro, couple rotating mass mechanically to the grid and provide inertia as an inherent by-product of generation. A sudden imbalance is therefore resisted by the kinetic energy stored in rotating shafts, slowing the rate at which frequency changes. By contrast, modern wind, solar and battery assets are normally connected through power electronics, so they do not naturally provide the same inertial and fault-current characteristics unless equipped with appropriate controls, such as synthetic-inertia or grid-forming capability (Milano et al., 2018; ENTSO-E, 2020; NESO, 2025j). As fossil plants retire, stability services that were previously bundled with synchronous generation must therefore be replaced either by dedicated equipment, such as synchronous condensers and voltage-support devices, or by new controls on inverter-based resources, including grid-forming inverters (NESO, 2023; ENTSO-E, 2020).

These systems operate across several timescales. Inertia and fault-current contribution act almost instantaneously; fault ride-through and dynamic voltage support operate over milliseconds to seconds; fast frequency response acts within seconds; primary and secondary frequency response operate over seconds to minutes; and reserves and dispatch act over minutes to hours. Some are grid-code capabilities, some are network stability requirements, and some are balancing-market services. The distinction matters because each timescale requires different technical capabilities and different procurement mechanisms. Great Britain and Germany have made different institutional choices about how to secure these capabilities as the fossil fleet retires, a difference developed in Chapter 6 below.

5. Quantifying the bottleneck: constraint costs and curtailment

5.1 Great Britain: £2.7 billion in balancing costs, £1.7 billion in thermal constraints

In the financial year 2024/25, NESO spent approximately £2.7 billion balancing the Great British electricity system, up ten per cent on the previous year. Thermal constraints were the dominant driver, accounting for £1.7 billion of the total, with their volumes up eighty-one per cent year-on-year to 13.5 TWh (NESO, 2025h). Northern Scotland remained central to the constraint problem: NESO attributes the 2024/25 increase partly to planned outages in Scotland, high wind output and reduced transfer capacity across key Scottish constraint boundaries (NESO, 2025h). Nationally, thermal constraint costs have nearly trebled since 2020/21, and NESO projects balancing costs could peak at around £8 billion in 2030 unless critical network projects are brought forward, with accelerated delivery able to avoid up to around £4 billion of costs in that year (NESO, 2025h).

Constraint costs arise from a two-sided balancing action. When generation exceeds boundary transfer capacity, NESO pays generators in the exporting region to reduce output through a "bid" payment and simultaneously pays generators in the importing region to increase output, usually at a higher "offer" price. The net cost is then recovered from consumers across Great Britain through the Balancing Services Use of System (BSUoS) charge, even though the underlying physical problem is geographically concentrated. Analysis by Regen (2024) shows that approximately seventy-six per cent of 2024 constraint costs reflected payments to gas generators to increase output, following "offers", and roughly twenty-four per cent reflected payments to wind generators to curtail. Hence the cost is not primarily a payment to wind farms, but a fuel and premium payment to gas plants dispatched out of merit order to replace constrained wind.

5.2 Germany: €2.8 billion in 2024

German redispatch volumes and costs follow a similar pattern. In 2024, total congestion-management volumes reached 30,304 GWh and preliminary total costs were €2.78 billion, down from €3.34 billion in 2023 largely because of lower gas and wholesale prices rather than a fundamental improvement in network capacity (SMARD, 2025). Of the adjusted generation, 22,777 GWh came from market-based dispatchable plants and 9,374 GWh from renewable curtailment, the latter down eleven per cent year-on-year. Offshore wind curtailment nevertheless rose sharply in the fourth quarter after a prolonged December wind event exceeded transport capacity (SMARD, 2025).

The federal-state breakdown shows that the most-curtailed onshore wind states in 2024 were Niedersachsen (912 GWh) and Schleswig-Holstein (806 GWh), precisely the northern states whose output the SuedLink and A-Nord corridors are designed to evacuate south (SMARD, 2025). Paralelly, solar curtailment nearly doubled in 2024 to 1,389 GWh, concentrated in Bavaria (986 GWh). This points to the growing distribution-network dimension of congestion as solar capacity rises. Of total renewable curtailment, roughly seventy-four per cent was caused by transmission-level bottlenecks, down from eighty per cent in 2023, while the distribution-level share rose to twenty-six per cent (SMARD, 2025).

5.3 The mechanics of curtailment: what happens when NESO tells a wind farm to reduce output

Curtailment is often described as "wind farms being switched off", but the actual mechanism is more nuanced and carries important economic consequences. In Great Britain, NESO uses

the Balancing Mechanism to manage real-time imbalances and network constraints. Ahead of each settlement period, Balancing Mechanism Units submit Bids, indicating the prices at which they are willing to reduce output or increase demand, and Offers, indicating the prices at which they are willing to increase output or reduce demand (Elexon, n.d.; NESO, 2025k). When forecast flows across a boundary exceed operational limits, NESO's control room instructs curtailment by accepting the Bids of generators in the exporting region, typically Scottish wind, while simultaneously accepting Offers from generators in the importing region to replace the lost energy.

Crucially, settlement is pay-as-bid rather than at a market-clearing price. A wind generator instructed to turn down will normally seek compensation for lost marginal revenue, net of any avoided costs. For subsidised generators, this may include additional foregone Renewables Obligation or Contracts for Difference-related revenue, because support payments depend on actual metered output. Curtailment payments therefore embed compensation for foregone market and support revenue, not just the physical act of switching turbines down, which helps explain why they have become politically contentious (Ofgem, 2024b; Regen, 2024; Intini et al., 2023). To limit the exploitation of local market power during transmission constraints, Ofgem applies the Transmission Constraint Licence Condition, which prohibits generators from obtaining excessive benefit from bid prices during transmission-constraint periods (Ofgem, 2024b).

On the replacement side, Offers from gas generators in England and Wales typically reflect fuel costs, start-up and ramping costs, short-run opportunity costs and scarcity premia. Unlike constrained wind Bids, these Offers are not primarily constrained by lost subsidy revenue; they are shaped by wholesale conditions and Balancing Mechanism competition. The structural consequence is that gas generators capture the bulk of constraint payments as shown earlier and this asymmetry reflects the higher cost of replacing constrained wind with out-of-merit thermal generation.

Germany's redispatch regime works on a different principle. Under Redispatch 2.0 and §13a EnWG, the system does not rely on the same pay-as-bid balancing mechanism for congestion management. Instead, network operators identify redispatch measures using standardised processes based on effectiveness and costs, and compensation is calculated administratively rather than through practically unconstrained strategic bidding (Bundesnetzagentur, 2025b). Renewable operators are compensated under rules that account for foregone generation and applicable support arrangements, while conventional generators are compensated through

regulated cost-based arrangements. The economic consequence is that generators have less scope to monetise local market power through congestion bids. The two regimes therefore generate sharply different incentive dynamics: GB's pay-as-bid system rewards bidding strategy and creates the political optics of "wind paid to switch off", while Germany's cost-based system suppresses both gaming opportunities and headline payment figures, but at the price of weaker locational signals to investors. A market-based redispatch has been proposed for Germany's Redispatch 3.0 and remains contested on efficiency versus gaming-risk grounds (Hirth and Schlecht, 2019; Hirth et al., 2019).

Whichever regime applies, curtailment is not in itself evidence of system failure. A fully renewable-dominated system with zero curtailment would require a grid so over-built that the marginal line almost never carries flow, which would be economically inefficient. The relevant question is whether observed curtailment is above the level that would be efficient relative to the cost of additional reinforcement, storage and flexibility (Gill and Bell, 2024; UKERC, 2025). In Great Britain, NESO reports 13.5 TWh of thermal-constraint volumes in 2024/25, while separate market analysis suggests that around one-tenth of GB wind output was curtailed in 2024, against UK wind generation of 83.3 TWh (NESO, 2025h; DESNZ, 2025a; Financial Times, 2025). This is consistent with an acute network bottleneck. In Germany, renewable curtailment of 9.4 TWh against roughly 260 TWh of renewable generation imply a much lower aggregate curtailment intensity, (SMARD, 2025). Both numbers point in the same direction as the institutional comparison: GB's binding corridors produce acute wind curtailment and high replacement costs, while Germany's wider redispatch regime spreads congestion management across a broader set of assets and network levels.

5.4 Where the GB bottleneck sits: boundary-level stress and cost concentration

The GB bottleneck appears concentrated on a small number of specific corridors. To test this, I constructed a boundary-level stress indicator using two NESO datasets: the ETYS March 2025 GIS shapefile, which provides polygon geometry for the named transmission boundaries, and the Day Ahead Constraint Flows and Limits dataset, which records half-hourly Flow (MW) and Limit (MW) values for operational constraint groups from October 2019 to April 2026 (NESO, 2025f; NESO, 2025g). For each operational constraint group, I computed the ratio of observed day-ahead flow to binding capacity limit at each half-hour, then derived mean stress, near-saturation frequency, defined as $\text{Flow/Limit} > 0.9$,

and binding-breach frequency, defined as $\text{Flow/Limit} > 1.0$. This was mapped onto ETYS boundary polygons where naming aligned; remaining constraint groups are subregional or intertrip zones that do not correspond to a single ETYS polygon but lie within Scotland or the Anglo-Scottish corridor.

The pattern is stark (Table 1). Of the £520m in thermal constraint cost attributable to specific boundaries between 22 April 2024 and 31 March 2025, boundary B4 alone accounts for £305m hence 58.7% of the total despite being only one of 23 operational constraint groups. B4, B6 and B7a together accounted for over 90% of thermal constraint cost, and all three sit on the Scotland–England corridor. Mean stress on Scottish and Anglo-Scottish boundaries averages 0.65, against 0.35 for boundaries in England and Wales. Scottish and Anglo-Scottish boundaries breach their binding limit more than seven times as frequently. The concentration of stress and cost therefore confirms that GB’s constraint problem is not a diffuse system-wide lack of capacity, but an acute corridor problem concentrated around Scotland and the Anglo-Scottish boundary.

Three analytical consequences follow. First, the "build more wires" debate resolves into a narrower question: the specific corridor reinforcements that matter most are the Eastern Green Link subsea HVDC projects (EGL1–4), whose operational dates of 2029 to 2033 therefore dominate the forward-cost trajectory. Second, the concentration explains NESO's development of corridor-specific instruments such as the B6 Constraint Management Pathfinder and the Local Constraint Market, both targeted at managing constraints around the B6 boundary (NESO, 2025m; NESO, 2025n). Third, the B4–B6–B7a corridor crosses three transmission owners (SSEN, ScottishPower and National Grid), which might convert this physical engineering problem into an institutional coordination problem at precisely the point where the costs are largest.

Table 1. Summary table of boundary stress and thermal constraint costs, 22 April 2024–31 March 2025

Boundary	Group	Total cost (£m)	Share of total cost (%)	Average stress	Average daily p95 stress	Half-hours >0.9 (%)	Half-hours >1.0 (%)
B4	SSE-SP	305.2	58.7	1.12	1.73	52.8	46.6
B6	SCOTEX	101.7	19.5	0.87	1.31	39.8	34.1
B7a	SSHARN	62.5	12.0	0.88	1.16	44.0	33.9

B15	ESTEX	46.2	8.9	0.48	0.62	1.0	0.0
LE1	SEIMP	4.7	0.9	0.36	0.63	4.2	2.3

Note: Stress is calculated as day-ahead Flow/Limit. Cost shares refer only to the selected thermal constraint-cost boundaries included in the table, not to all GB balancing costs or all thermal constraint costs.

5.5 Convergent symptoms, divergent numbers

Adjusted for system size, balancing and redispatch costs per MWh generated are in the same order of magnitude in both countries. The more interesting pattern is directional. In both systems, costs have risen sharply since 2019 even as underlying wholesale prices have fluctuated, indicating a structural increase in congestion-management activity (NESO, 2025h; SMARD, 2025). In both, costs are forecast to rise further before falling as major transmission reinforcements commission around 2028–2032. And in both, the costs are socialised across all consumers rather than paid by the beneficiaries of curtailment or the regions whose demand drives replacement dispatch.

The socialisation of constraint costs is one symptom of a deeper pattern: the bottleneck is not bounded by the annual constraint bill but extends, as a shadow cost, across the pipeline of new generation and new demand waiting to connect. In Great Britain, the previous first-come, first-served connections model led to more than 700 GW of generation and storage projects waiting for grid access, around four times the capacity required for 2030, and left some shovel-ready projects waiting up to a decade (NESO, 2025a). The queue inflation was itself a consequence of physical scarcity and first-come, first-served incentives: developers applied defensively for early slots because connection dates had become slow and uncertain, while speculative and immature projects further delayed projects ready to build. NESO’s December 2025 connections reform reset the generation-and-storage pipeline to 283 GW of projects meeting readiness and strategic-alignment criteria, formalising the diagnosis that the queueing mechanism had become part of the bottleneck (NESO, 2025a). Germany faces a parallel dynamic from a different direction: of the roughly 16,800 km of transmission lines planned to relieve chronic north–south congestion, only a fraction had been completed by mid-2025, leaving generators in the wind-rich north and loads in the industrial south exposed to limited transfer capacity and multi-billion-euro annual redispatch costs (Bundesnetzagentur, 2026a; SMARD, 2025).

The same upstream effect now appears on the demand side, through a different mechanism: saturated local transmission capacity is delaying connections for major loads, with some data-centre developments in West London facing waits of up to a decade (London Assembly, 2025). Whether the binding constraint is queue congestion for new generators or physical saturation for new loads, the result is that location-flexible industrial consumers such as some data-centre operators, electrolyser developers and reshored manufacturers increasingly treat grid-connection timelines as a first-order siting criterion. The transmission constraint is therefore beginning to shape industrial geography, not just electricity-system costs.

Part III — Divergent institutional responses

6. Market design and price signals

6.1 Mechanism: uniform pricing and the bill components it lives inside

Both Germany and Great Britain operate national wholesale bidding zones: Great Britain retains a single GB-wide wholesale electricity market, while Germany continues to operate within the German–Luxembourg bidding zone (DESNZ, 2025; BMWEL, 2025; ENTSO-E, 2025). The choice of this pricing geography, a single zone rather than multiple price areas reflecting locational scarcity, weakens the signal that the bottleneck exists. Because the wholesale price a Scottish wind developer faces is identical to the price an English developer faces, the signal hides the fact that Scottish output may require scarce north–south transmission capacity to reach demand.

This design simplifies trading and reduces basis risk for generators, but produces a wholesale price signal that does not reflect local transmission scarcity when the grid is congested. During constraint periods, the shadow marginal cost of delivered electricity differs sharply between regions. In an exporting region, additional renewable output may have low delivered value if it cannot cross the constraint; in a deficit region, the marginal MWh may require out-of-merit replacement generation. Under this uniform national pricing, these costs later appear as balancing costs, constraint costs and network charges. In other systems like locational marginal pricing (LMP, used in most US ISOs), the same physical reality would appear as different nodal prices reflecting congestion and losses. Under zonal pricing, as used in markets such as the Nordic region and Italy, it would appear as different prices between adjacent bidding zones (ISO New England, 2024; Pollitt, 2023; Eicke and Schittekatte, 2022).

The wholesale price is only one channel through which grid scarcity reaches consumers. The next section therefore separates wholesale-price signals from network and balancing charges, where hidden congestion costs are recovered.

6.2 Distribution: who pays, through which components

The distributional consequences of uniform-zone pricing in GB are visible only when consumer bills are decomposed. The wholesale energy charge, which represents roughly 30 to 40 per cent of a typical residential bill, is set at a single national clearing price. TNUoS charges are regionally differentiated by region. DUoS charges set by regional DNOs also vary by area. BSUoS charges, which include the socialised costs of balancing and constraint actions, were levied on both generators and demand until April 2023 but following CMP308 they are now recovered solely from final demand through an ex-ante fixed tariff (NESO, n.d.-b). The result is a mixed structure: regional network charges partially reflect location, but the wholesale price remains national and balancing costs are socialised through demand-side BSUoS.

Germany's distributional pattern is structurally analogous but operates through different components. The wholesale price is uniform within the Germany–Luxembourg bidding zone, and Germany's 2025 bidding-zone action plan confirms the decision to retain this single-zone approach (BMWE, 2025; ENTSO-E, 2025). Transmission grid charges were harmonised across the four TSOs in 2023, so that all consumers nationally face the same TSO-level component, but distribution-level network charges set by approximately 880 DSOs continue to exhibit substantial regional variation. Since 2025, the §19 StromNEV redistribution mechanism socialises a portion of the cost burden of renewable-heavy regions through a uniform 1.558 ct/kWh surcharge across all consumers, and the €6.5 billion federal subsidy that took effect in January 2026 further dampens transmission charges (Bundesnetzagentur, 2024a; Netztransparenz, 2025; Bundesregierung, 2026). Southern industrial consumers in Bavaria and Baden-Württemberg therefore benefit from a national wholesale price that averages in cheaper northern wind, while the cost of physically delivering that electricity is recovered through grid charges that remain partly regional but are increasingly redistributed through national mechanisms.

This bill-component structure is itself a partial answer to the efficiency-equity tradeoff Okun (1975) identified: zonal pricing maximises allocative efficiency by letting prices reflect true

local marginal costs, while uniform pricing sacrifices some of that efficiency in exchange for distributional control, leaving regional fairness to be addressed through separate instruments such as targeted network-charge rebates, direct fiscal transfers or regional industrial policy. Germany has long operated under this second model while GB confirmed the same approach via the July 2025 REMA decision (DESNZ, 2025; Bundesnetzagentur, 2024).

6.3 The theoretical case and the empirical evidence

The economic theory of locational pricing is well-established. Schweppe et al. (1988) laid out the foundational framework for spot pricing electricity, and Hogan (1992) developed the financial transmission rights extension that underpins most US ISO markets. Joskow (2007) summarises the theoretical case: when transmission is congested, marginal cost varies by location, and a uniform price distorts both dispatch (wrong generators run) and investment (generators and loads locate in the wrong places). These distortions become more important in renewable-heavy systems, where congestion is increasingly structural rather than occasional. The theoretical case for zonal or nodal pricing in GB and Germany has been made in detail by Eicke and Schittekatte (2022) and Pollitt (2023), who argue that the status quo systematically weakens locational signals and can over-reward generation located far from demand.

The empirical case in GB rests heavily on modelling produced during the REMA debate. FTI Consulting's 2025 analysis for Octopus Energy estimated substantial consumer and welfare benefits from zonal pricing, driven by lower constraint costs, more efficient siting and reduced need for transmission investment (FTI Consulting, 2025a). A further FTI/Octopus analysis claimed that zonal pricing could avoid nearly 3,000 km of new power lines and save consumers up to £27 billion by reducing unnecessary grid build (FTI Consulting, 2025b; Octopus Energy, 2025). Counter-modelling commissioned by SSE argues that these benefits fall sharply once NESO's Beyond 2030 network plans are included, and that even small increases in the cost of capital could eliminate the modelled gains (LCP Delta, 2024; Frontier Economics, 2024). Whether the locational-elasticity and investment-risk assumptions in the competing models are realistic is therefore an open empirical dispute, explored further in Chapter 8..

6.4 Empirical resolution: the REMA decision and the German bidding-zone rejection

In July 2025, the UK government announced that it would retain the single GB-wide wholesale market and reject zonal pricing reform (DESNZ, 2025; Norton Rose Fulbright, 2025). The decision was justified on the grounds that retaining national pricing would provide investor certainty, avoid the implementation risks associated with a major market-design change, and allow the government to pursue locational signals through a reformed national-pricing package with stronger siting and operational signals rather than through wholesale-price separation (DESNZ, 2025). The adopted alternative, Reformed National Pricing, relies on a Strategic Spatial Energy Plan due by end-2026, TNUoS reform by 2029 at the latest, and balancing-mechanism reforms including shorter settlement periods (DESNZ, 2025; Drax, 2025; HSF Kramer, 2025; NESO, 2025).

Germany's approach was similarly conservative, but the timeline should be stated differently. The 2025 ENTSO-E Bidding Zone Review found that splitting Germany–Luxembourg into five bidding zones produced the highest positive economic-efficiency result among the analysed German configurations, but the German government chose to retain the single bidding zone and pursue an action plan focused on grid expansion and congestion management instead (ENTSO-E, 2025; BMW, 2025). The explicit justification was investor continuity, European market integration and the view that network reinforcement should address structural congestion; the underlying politics was the recognition that zonal pricing would create a visible south-pays-more outcome unacceptable to Bavaria and Baden-Württemberg (Monopolkommission, 2023; Clean Energy Wire, 2025).

Both governments therefore arrived at the same destination through different institutional routes: national wholesale pricing, with locational signals pushed into network charges, planning and balancing reforms. GB's decision emerged from public consultation; Germany's from federal and European market-governance negotiation. The common outcome suggests that investor-certainty arguments and regional distributional politics can preserve uniform-zone pricing even when the theoretical efficiency case points the other way.

7. Grid ownership, governance and investment

7.1 Governance architecture

Great Britain and Germany have arrived at structurally different transmission-governance configurations, but the difference is finer than the "single TSO versus four TSOs" framing

implies. In Great Britain, system operation and asset ownership are institutionally separated. The National Energy System Operator (NESO), spun out of National Grid in October 2024, is a publicly owned corporation responsible for system operation, network planning and balancing-mechanism procurement (DESNZ, 2024; NESO, 2024f). The transmission assets remain privately financed regulated monopolies owned and operated by National Grid Electricity Transmission, SP Energy Networks, Scottish Hydro Electric Transmission and SSEN Transmission, which all mix private and public capital (Ofgem, 2024a). In Germany, ownership and operation sit together within four TSOs: TenneT DE, 50Hertz, Amprion and TransnetBW (Bundesnetzagentur, 2025a), whose ownership structures also mix public and private capital. Hence what differs structurally is whether system operation is unified (GB) or distributed across four federally-coordinated control areas (DE).

7.2 Regulatory incentives and the asset-bias question

The two regulatory frameworks share a common analytical lineage but treat capital expenditure differently. Great Britain operates under Ofgem's RIIO framework, now in its third iteration (RIIO-T3, April 2026 to March 2031), which sets revenue caps on a totex (total expenditure) basis with output delivery targets and efficiency incentives explicitly designed to soften the distinction between capital and operating expenditure (Ofgem, 2025a). Germany operates under Bundesnetzagentur revenue-cap regulation under the *Anreizregulierungsverordnung*, with capital expenditure recognition through capex mark-ups that bring qualifying network investments into allowed revenues on a project-specific basis (Bundesnetzagentur, 2025a). Both frameworks impose efficiency tests and benchmarking, and neither blindly favours "more steel in the ground." However, a residual Averch-Johnson risk (Averch and Johnson, 1962) remains: capital expenditure earns a regulated return on the Regulatory Asset Base, while non-asset solutions such as operational optimisation, congestion contracts and flexibility procurement do not build the asset base in the same way (Averch and Johnson, 1962). TenneT's own 2022 analysis acknowledges that European TSO benchmarking "leads to a possible preference for grid [reinforcement]" over operational alternatives (TenneT, 2022). The empirical pattern reviewed earlier, selective WAFB deployment in Germany and still limited non-asset procurement in Great Britain despite repeated NESO commitments, is consistent with asset bias remaining a feature of both regimes.

7.3 Strategic corridor cases: SuedLink and the Eastern Green Links

The investment programmes are large and broadly proportional to system size. NESO's Beyond 2030 plan proposes around £58 billion of transmission investment to 2035 (NESO, 2024b), while Germany's Netzentwicklungsplan 2037/2045 estimates €360–390 billion of total transmission expansion costs by 2045 across scenarios (Netzentwicklungsplan, 2025). Within these portfolios, the strategic-corridor megaprojects matter disproportionately because Chapter 5 showed that binding constraint costs are concentrated on a small number of north-south corridors. AC remains the dominant technology for inland regional reinforcement, meshed networks and shorter-distance projects, while HVDC is chosen where things such as distance, subsea routing, controllability and transfer scale dominate (NESO, 2024b; IEA, 2025). The Anglo-Scottish corridor is being addressed by four Eastern Green Link (EGL) subsea HVDC projects, each rated at 2 GW, with construction beginning in 2025 and commissioning staggered between 2029 and 2033 (SEN Transmission, 2025; Iberdrola, 2026). Germany's primary north-south corridor reinforcement is SuedLink, two 2 GW underground HVDC lines totalling around 700 km from Schleswig-Holstein to Bavaria and Baden-Württemberg, expected to commission in 2028 (Bundesnetzagentur, 2025c). These are strategic corridor cases, not proxies for total grid expansion: they share function (north-south evacuation of peripheral wind to central demand), transfer scale (2 GW per circuit) and timing (late-2020s to early-2030s commissioning). The Anglo-Scottish corridor and the German north-south corridor are precisely the points at which the binding constraint costs documented in Chapter 5 are concentrated. They are therefore valid as comparative cases of how each system delivers its most strategically important reinforcements, while not licensing inferences about the average commissioning speed of all transmission investment.

7.4 Consent stacks and the supply-chain equaliser

Both systems operate multi-step approval stacks. In Great Britain, EGL2 required Ofgem Final Needs Case approval (July 2022), Aberdeenshire Council planning permission for the converter station (May 2022), Marine Scotland and Marine Management Organisation marine licences (2023), East Riding Council planning permission for onshore cables (May 2023), a Compulsory Purchase Order confirmation by the Secretary of State (February 2025) and Ofgem final ASTI approval (August 2024) before construction could begin (Ofgem, 2022; Eastern Green Link 2, 2024a; Ofgem, 2024b; DESNZ, 2025). In Germany, SuedLink required federal corridor planning, section-by-section planning approvals, with the final approval issued in October 2025, and BNetzA-granted early-start permits enabling phased

construction from 2022 onwards (Bundesnetzagentur, 2025c). The institutional shapes of the consent stacks differ, but neither is single-stage. This matters because it weakens a tempting causal explanation: GB's relative speed on early east-coast subsea projects mostly reflects the fact that subsea routing reduces exposure to continuous land-based route consent.

A common marginal constraint in both systems is now the European HVDC supply chain. The IEA reports that cables take two to three years to procure, large power transformers up to four years, and some HVDC cables more than five years (IEA, 2025). The European converter-station market is concentrated around a small number of major suppliers, including Hitachi Energy, Siemens Energy and GE Vernova. TenneT's 2023 2 GW offshore programme converted part of this exposure into a portfolio-scale framework, with around €30 billion of contracts covering 14 offshore grid connection systems due to be realised by 2031, while GB has accelerated strategic transmission procurement through Ofgem's ASTI framework and project-specific early construction funding (Offshore Engineer, 2023; Ofgem, 2024b; Ofgem, 2025b). Both responses are recognisably responses to the same external bottleneck: differently governed systems are now exposed to a common European supply-chain constraint that domestic institutional reform can mitigate through early procurement, but cannot remove directly. This is the strongest comparative finding of the chapter and the empirical bridge to Chapter 8: when supply-side institutional acceleration runs into externally set physical limits, demand-side instruments stop being optional and start being analytically central.

8. Demand-side levers and the limits of spatial flexibility

8.1 Definition and analytical framework

The IEA defines demand flexibility as "the ability to adjust the timing or amount of electricity use in response to system needs" (IEA, 2025b). The definition matters because it identifies what demand-side instruments physically do: they shift consumption in time or volume. For this dissertation, the definition can be extended analytically to distinguish temporal, volumetric and spatial flexibility. Temporal flexibility, such as rescheduling EV charging, smoothing industrial load profiles or dispatching battery storage, can reduce peak stress and absorb surplus renewable generation, thereby lowering the marginal cost of constraint events (IEA, 2025b; DESNZ, NESO and Ofgem, 2025). However, it only relieves a specific transmission boundary if the flexible demand is located on the relevant side of the

constraint. Spatial flexibility, by contrast, means siting new electricity-intensive loads closer to renewable resources. It directly addresses the peripheral-to-central transfer problem identified in Chapter 5, but is scarcer because most demand is geographically inelastic, at least in the short run: heat pumps follow buildings, EV charging follows journeys, industrial heat follows industrial sites and low-latency data centres need to be close to populated areas.

Other analytical axes include demand reduction, such as efficiency improvements, retrofits and modal shift; distributed generation; and smart-grid infrastructure, including smart metering, half-hourly settlement and aggregator interfaces. They all reduce transmission burden distinctly through reducing aggregate consumption, reducing net demand drawn from the transmission network, and enabling the operational dispatch of flexibility respectively (UKERC, 2025; Bundesnetzagentur, 2026; Fraunhofer ISE, 2026). Germany illustrates the distributed-generation channel: installed solar PV capacity reached 116.8 GW by end-2025, while self-consumed PV reached 12.28 TWh in 2024, reducing the volume drawn from or injected into the public grid in some periods (Fraunhofer ISE, 2025; Fraunhofer ISE, 2026). The chapter touches on each in passing rather than treating them as coordinate categories, because the spatial/temporal cut maps most directly onto Chapter 5's peripheral-to-central transfer problem.

8.2 Quantitative anchors: what is on the table by 2030

Great Britain's Clean Flexibility Roadmap, published jointly by DESNZ, NESO and Ofgem in July 2025 sets a 2030 target of 10–12 GW of consumer-led flexibility by 2030 within a total system flexibility envelope of 51–66 GW including grid-scale batteries, interconnectors, long-duration storage and dispatchable low-carbon supply (DESNZ, NESO and Ofgem, 2025). Demand-side instruments alone therefore represent roughly one-fifth of total system flexibility on the 2030 horizon. The Demand Flexibility Service provides the operational architecture: in winter 2024/25 NESO received 10.98 GWh of DFS bids and accepted 5.45 GWh across 44 events, with 1.98 million MPANs registered by the end of March 2025 (NESO, 2025p). Germany has no comparable unified 2030 target. FfE's expert opinion for the four TSOs estimated positive load management potential at 1.2 GW in 2019, provided mainly by the aluminium, paper and chemical industries, rising in scenario projections to 13.5 GW by 2045, with industrial cross-section technologies and the tertiary sector overtaking traditional industrial processes from 2035 onwards (FfE, Consentec and Fraunhofer ISI, 2021; FfE, 2022). The 2030-horizon figure scales to approximately 5 GW. Germany's recent

flexibility-related reforms including the §14a EnWG controllable-load provisions in force since January 2024 and mandatory dynamic electricity tariffs from January 2025 are sectorally targeted rather than aggregated under a single national flexibility target (Bundesnetzagentur, 2023; Bundesnetzagentur, 2025a). Demand-side flexibility is therefore more explicitly organised in Great Britain than in Germany.

8.3 The spatial-flexibility limit: electrolyzers as case study

The cleanest test of whether demand-side measures can address spatial mismatch is the German electrolyser fleet, because hydrogen production is among the few large new loads that is, in principle, spatially mobile and is being deliberately planned as a grid-supportive technology. The Netzentwicklungsplan 2037/2045 assumes that electrolyzers will be sited where they relieve transmission bottlenecks and absorb otherwise-curtailed renewables (Transmission System Operators, 2023). This, however, is a planning assumption rather than a guaranteed market outcome. FfE modelling published in January 2025 provides the cleanest counterfactual evidence: under uniform German pricing and without a minimum expansion target, the model deploys no electrolysis in Germany by 2030; under a north–south bidding-zone split, 16 GW deploys in the north as early as 2030, driven by lower residual load in the northern zone (Helmer and Ganz, 2025). The implication is not simply that electrolyzers will locate in the north, but that they do so only when the market design gives them a reason to internalise the system value of northern siting.

The current German connection pipeline reinforces this ambiguity. Germany’s national hydrogen strategy targets at least 10 GW of domestic electrolyser capacity by 2030, while the TSOs already face a much larger pipeline of electrolyser connection requests, concentrated around northern renewable and offshore-grid regions (BMWK, 2023; Netztransparenz, n.d.). That concentration could be beneficial if it absorbs renewable output that would otherwise be curtailed. But it could also create new local connection and stability problems if large electrolyzers cluster at already-stressed substations or disconnect simultaneously during faults, which is why the German TSOs have developed specific extra-high-voltage connection requirements for electrolyser facilities (Netztransparenz, n.d.; VDE, 2025). Electrolysers therefore show the central limit of spatial flexibility: it is technically plausible and systemically valuable, but it does not emerge automatically from uniform pricing and without locational signals and coordinated connection rules, the same flexible loads modelled as relieving the constraint can become another source of grid stress.

8.4 Three barriers to scaling

Three barriers consistently emerge across the GB and German evidence base. First, price signals are misaligned. First, price signals remain misaligned. As Chapter 6 showed, both systems retain uniform wholesale pricing and rely on network charges or redistributive mechanisms to manage locational tensions. That architecture weakens the business case for spatial flexibility: electrolyzers, data centres and other mobile loads are not automatically rewarded for locating where they reduce congestion (DESNZ, 2025; Bundesnetzagentur, 2024; Bundesregierung, 2026).

Second, market access and metering remain binding. NESO has identified access to markets, service stacking, balancing-mechanism participation and aggregator participation as priority reform areas under its Demand-Side-Flexibility Routes to Market Review (NESO, 2025c), indicating that technically available flexibility is still not fully able to reach the markets where it would be valued (NESO, 2024; NESO, 2025q. In Germany, the bottleneck is more infrastructural: mandatory smart-meter rollout has progressed slowly: it reports approximately 640,000 mandatory installations by end-2024, equivalent to a 13.9 per cent equipment rate, against approximately 53 million metering points nationally (FfE, 2025; Bundesnetzagentur, 2024b).

Third, what has been less covered in this dissertation but relevant in this chapter is that congestion is increasingly shifting to distribution networks, which makes flexibility harder to scale because the relevant constraints become local, fragmented and feeder-specific. As distributed solar PV, EV charging and heat pumps proliferate at the low-voltage level, the binding constraint is no longer only whether bulk power can move across national transmission corridors, but whether local networks can absorb injections and serve new loads at specific points. German DSOs project network investment requirements of approximately €110 billion by 2033 and €207 billion by 2045 (Bundesnetzagentur, 2025a). Hence, this make demand-side flexibility more complex: Flexibility must increasingly defer, target and optimize local reinforcement across many distribution networks.

8.5 Implication and handoff

The strongest defensible claim is that demand-side measures complement transmission reinforcement on the binding 2030 horizon. The longer-horizon literature is more optimistic: Li and Pye (2018) estimate that demand-side flexibility only from smart appliances and

passenger EVs can reduce UK peak load by approximately 7 GW (9 per cent) and cumulatively save £30.9 billion by 2050; UKERC and Piclo modelling suggest demand-side measures could reduce long-run UK network investment requirements by up to 50 per cent on the same 2050 horizon (Li and Pye, 2018; Piclo, Element Energy and Graham Oakes, 2020; Kuzemko et al., 2025). The frontier question is therefore whether the political and regulatory architecture can deliver the targeted demand-side capacity fast enough to matter, a political-economy question about cost visibility, narrative attribution and the durability of the overall settlement, taken up in Chapter 9.

9. Political economy and social acceptance

This chapter steps back to ask why some cost arrangements survive politically and others do not, and how the visibility of costs shapes the institutional configurations each country has settled into. The argument is that the political durability of British and German grid-policy outcomes rests less on their economic efficiency and justice than on whether their costs are individually observable to consumers and voters.

9.1 Mechanism: visible versus invisible costs

Great Britain has created explicit markets for stability services that were previously free by-products of synchronous fossil generation. The NESO Stability Pathfinder contracts synchronous condensers, hydro plants and grid-forming batteries to provide inertia and short-circuit level through fixed availability payments; the Dynamic Containment, Dynamic Moderation and Dynamic Regulation services contract batteries and other fast-response providers for frequency response; the Obligatory Reactive Power Service compensates generators for voltage support (NESO, 2025e). Total contracted spend across these stability and response markets ran to approximately £200 million in 2024/25 and is reported in NESO's Annual Balancing Services Spend Report in itemised form (NESO, 2025e).

Germany relies more heavily on mandatory hardware standards embedded in grid-connection rules which require new generators to provide dynamic voltage support, fault ride-through, frequency-dependent active power response and reactive power capability as a condition of connection (VDE, 2023; VDE, 2018). These domestic standards transpose the European Network Code on Requirements for Generators (Commission Regulation (EU) 2016/631), with grid-forming and inertia-like capability now being incorporated through forthcoming amendments (FNN, 2025). Developers must design and pay for inverters and controls to meet these standards, so costs are absorbed into project capex rather than paid through an explicit

service market. The economic effect is not necessarily different as developers recover these costs through EEG and CfD auction bids ultimately borne by consumers, but the visibility is dramatically different: No single published line-item says what households are paying for inertia, fault ride-through or grid-forming capability.

9.2 Distribution: backlash, transfers and political sustainability

The political durability of the two cost arrangements differs less because of whether costs are technically visible than because of whether they can be attributed to a politically salient recipient. In Great Britain, constraint costs were politically attached to a specific actor: the narrative that "wind farms are being paid to switch off" gave the cost a transfer recipient and a target for opposition. This narrative was politically effective during the REMA consultation, although it was a factual inaccuracy, and it contributed to the political environment in which DESNZ ultimately rejected zonal pricing in July 2025 (DESNZ, 2025; Norton Rose Fulbright, 2025).

Germany's redispatch costs are passed through to consumers via grid charges (Netzentgelte) that are itemised on bills and exhibit substantial regional variation. They are therefore not invisible but what they lack is an equivalent narrative attribution: the redispatch mechanism's administrative cost-based compensation (Section 5.3) does not produce identifiable rent recipients in the way the GB market-bid mechanism does, and the political framing of grid charges has remained focused on grid expansion and network operator costs rather than on transfers to specific generators. The German 2023 cancellation of the planned €5.5 billion subsidy following the Constitutional Court's Climate and Transformation Fund ruling, and the eventual €6.5 billion subsidy that took effect in January 2026, were debated as fiscal-policy and industrial-competitiveness questions rather than as energy-transfer questions (Clean Energy Wire, 2023; Bundesregierung, 2026). Hence, the distributional structure is broadly similar to GB's but the political experience is "calmer" because the costs do not attach to an identifiable counterparty.

9.3 The coal-exit dimension

Germany's coal phase-out, currently legislated for 2038, sits at the intersection of four dimensions: bulk electricity supply, grid security, regional economic policy and electoral politics. Coal and lignite together produced 95.3 TWh of net public electricity generation in 2024, around 22 per cent of the public mix (Fraunhofer ISE, 2025). Coal and lignite plants also provide synchronous system characteristics, including real power, reactive power,

inertia, fault current and voltage support. In southern Germany, where the retirement of nuclear has removed a major source of synchronous generation, some conventional capacity is retained in the Netzreserve to support secure system operation while north–south reinforcements such as SuedLink are still being built. SMARD reports that reserve-plant operation rose to 1,788 GWh in 2024 from 1,214 GWh in 2023 (SMARD, 2025a).

The grid-related role of coal has three distinct components, and political discourse tends to conflate them. The first is narrowly operational: reserve plants dispatched outside the wholesale market when market-based redispatch is insufficient to relieve congestion. The second is market-based redispatch, where conventional plants, including gas, hard coal and lignite units, are instructed up or down to manage network constraints; in 2024, redispatch adjustments involving market-based dispatchable plants totalled 22,777 GWh (SMARD, 2025b). The third is residual coal generation that runs because Germany still lacks sufficient dispatchable replacement capacity, regardless of where the constraint sits. Grid expansion most directly reduces the second component and can also reduce part of the first by lowering reserve requirements: Bundesnetzagentur forecasts redispatch volume falling from around 25 TWh in the 2024/25 analysis to 17 TWh for 2025/26, partly because of grid-expansion progress (Bundesnetzagentur, 2025d). By contrast, replacing the synchronous services embedded in coal and lignite plants requires synchronous condensers, STATCOMs, grid-forming inverters or other stability-providing assets, while replacing residual baseload and dispatchable capacity requires gas or hydrogen-ready backup, storage of renewables and demand-side flexibility.

The political-economy contestation is concentrated in the eastern lignite regions, especially Lausitz, where coal-exit timing interacts with employment, regional identity and the €40 billion of structural-transition funding allocated under the Strukturstärkungsgesetz Kohleregionen (Bundesregierung, 2020; Oei et al., 2020). The western Rheinisches Revier has separately agreed an accelerated 2030 lignite exit with RWE, brought forward from the original 2038 horizon (BMWK, 2022). The stability argument is therefore invoked across several different problems, even though the technical case applies most directly to system security and reserve needs, while the regional transition problem follows a separate political timeline.

Great Britain's earlier coal exit, completed in October 2024, forced earlier and sharper solutions to all three problems: explicit service markets for stability, gas as the residual dispatchable replacement, and demand-side flexibility procurement through the Demand

Flexibility Service (HM Government, 2021). It also politically decoupled the stability question from regional economic-support politics, because the Scottish coal fleet had already been retired by the early 2020s and Welsh coal by 2020. Great Britain's politics of grid stability has therefore been primarily a politics of payments to private providers, not of regional industrial transition; Germany's remains both, and will continue to be until system-security questions and lignite-region transition questions are resolved on different timelines.

9.4 Planning, consent and the public

Both countries face local opposition to new overhead transmission lines routed through landscapes and communities that see limited direct benefit. Germany's 2015 decision to prioritise underground cables for major HVDC corridors, the Erdkabelvorrang, was a political-acceptance response to sustained opposition to overhead pylons, especially in Bavaria, despite the higher cost of undergrounding (BMWE, n.d.; Clean Energy Wire, 2026). undergrounding costs roughly six times more than equivalent overhead transmission depending on soil conditions, and the SuedLink investment is now estimated at €10 billion for a 700 km link, with commissioning pushed from 2022 to 2028 (Clean Energy Wire, 2023; Jacobs, 2022).

Great Britain's equivalent debate has centred on National Grid's Norwich-to-Tilbury project, a proposed 184 km, 520-pylon reinforcement between Norwich Main, Bramford and Tilbury, involving a new overhead line, opposed jointly by Norfolk, Suffolk and Essex County Councils on grounds of visual impact and landscape harm under the Holford Rules and the case for offshore or underground alternatives (National Grid, 2024; Suffolk County Council, 2024). Planning reform is ongoing: the Labour government's 2024 manifesto and King's Speech committed to streamlining the Development Consent Order regime, and the Planning and Infrastructure Bill introduced in March 2025 prioritises grid-connection projects that meet readiness and strategic-alignment criteria, revises National Policy Statements every five years, and shortens judicial review timelines following the Banner Review HM Government, 2025; NESO, 2025a).

The consenting timeline is consistently a binding constraint. In both countries, major transmission projects often take many years from identified need to energisation, and underground cable routes can add further planning, engineering and construction complexity. This physical-political timeline sits uncomfortably against the 2030 and 2045 targets and

helps explain why commissioning dates for major reinforcements cluster in the late 2020s rather than the mid-2020s, even where the need was identified much earlier. The IEA's observation that new grid infrastructure takes five to fifteen years to plan, permit and complete, compared with one to five years for renewable-generation projects, is a direct description of the German and British experience (IEA, 2023).

Part IV — Synthesis

10. Explaining the variation and the similarities

The comparison shows that Great Britain and Germany differ institutionally, but converge operationally. Great Britain has a unified system operator, a clearer national planning architecture, explicit flexibility markets and visible balancing charges. Germany has four TSOs, more federalised planning, stronger reliance on technical standards and grid-code obligations, and a larger role for network charges and fiscal redistribution. These differences shape how grid costs are generated, allocated and politicised. They determine who decides, who pays, how visible the cost is, and which actors become politically exposed.

Yet the central empirical finding is that these institutional differences do not fully explain delivery outcomes. Both systems face persistent congestion, large balancing or redispatch costs, and commissioning timelines clustered around the late 2020s and early 2030s. The explanation is that some binding constraints sit outside institutional design itself: planning consent, local opposition, HVDC converter and cable supply chains, skilled labour, and the physical mismatch between peripheral renewable generation and central demand. Institutions shape how systems adapt around these constraints, but do not remove them quickly.

This is the main source of variation. Great Britain has been faster at creating explicit markets and instruments: stability pathfinders, balancing services, the Demand Flexibility Service and a more articulated Clean Flexibility Roadmap. Germany has been stronger at embedding technical requirements and socialising costs through regulated network structures, but its demand-side flexibility architecture remains more fragmented. Conversely, Germany's redispatch regime reduces the political visibility of individual payments, while Great Britain's pay-as-bid balancing mechanism makes constraint costs easier to attribute to specific actors, even when the popular narrative is technically misleading.

The mutual lesson is therefore asymmetric. Germany could learn from Great Britain's clearer flexibility-market architecture and more explicit treatment of demand-side flexibility as an operational resource. Great Britain could learn from Germany's stronger integration of technical requirements into grid codes and its broader use of administrative cost-based redispatch to reduce gaming and political controversy. But neither model is simply superior. The right institutional solution is context-dependent: it depends on the geography of congestion, the political acceptability of regional price differences, the maturity of flexibility

markets, and the degree to which costs can be made visible without becoming politically toxic.

11. Conclusion, limits and policy implications

11.1 Conclusion

This dissertation asked why Great Britain and Germany continue to face large renewable-integration bottlenecks despite legally binding decarbonisation targets and historically large transmission-investment plans, and why two systems with different institutional architectures converge on similar delivery timelines and cost trajectories. The answer is that renewable integration is constrained not only by how transmission is governed, but by the interaction between physical geography, market design, supply chains, planning consent and political acceptance.

The core finding is that institutional architecture matters most for distribution, not speed. It affects whether costs appear as BSUoS charges, Netzentgelte, redispatch payments, fiscal subsidies or project capex; it affects whether the public blames wind farms, network operators, government or no one in particular. But it does not fully determine how quickly HVDC corridors, converter stations or new overhead lines can be delivered. The binding bottleneck is therefore not simply “too little grid investment”, but the limited capacity of institutions to convert investment plans into timely physical infrastructure.

11.2 Limits

The analysis has three limits. First, it focuses mainly on transmission, while distribution networks increasingly matter for solar PV, EV charging, heat pumps and local congestion. Second, the quantitative boundary-stress analysis is original for Great Britain but not symmetrically replicated for Germany, where the analysis relies more heavily on published redispatch indicators. Third, the dissertation does not conduct a formal welfare model of zonal pricing, demand-side flexibility or grid reinforcement. It compares institutional mechanisms and empirical indicators rather than estimating a full counterfactual cost-benefit ranking.

Future research could extend the boundary-stress method to Germany using spatial redispatch data, examine distribution-level congestion in both countries, or model how different mixes of HVDC expansion, demand-side flexibility and locational pricing affect constraint costs under alternative renewable-build scenarios.

11.3 Policy implications

The policy implication is not that grid expansion is unnecessary. It is that grid expansion alone is too slow to be the only answer. Governments should treat transmission reinforcement, demand-side flexibility, storage, locational signals and technical standards as complements rather than substitutes. Great Britain's priority should be scaling demand-side flexibility beyond pilot instruments and ensuring that Reformed National Pricing sends credible locational signals without reopening destabilising uncertainty. Germany's priority should be turning fragmented flexibility reforms into an operational architecture, while ensuring that network-charge redistribution does not erase incentives for efficient siting.

For both countries, the common lesson is that social acceptance depends on more than total cost. Costs must be visible enough to guide behaviour, but not so poorly attributed that they generate misleading backlash. The challenge is therefore to design institutions that reveal where congestion exists, reward actors who reduce it, and distribute the unavoidable costs transparently enough to remain politically durable.

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